

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1153

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. *(BL4)*

Assessment Tool Weightage: 40%

QUESTIONS:4

Total Marks:50

Annexure A

1. (Reg. No. 192411206)

- c) Develop a system using UML for implementing Stock Maintenance System. In this system, the customer can place orders and purchase items with the aid of the stock dealer and central stock system. These orders are verified and the items are delivered to the customer. Then the stock details are to be updated accordingly.
- d) Develop a system using UML for implementing Online Course Reservation System. This system is organized by the central management system. The student first browses and selects the desired course of their choice. The university then checks the availability of the seat if it is available the student is enrolled for the course.

UML Design and Implementation of Stock Maintenance System

1. Introduction

A Stock Maintenance System is an inventory management application that helps organizations maintain information about products, stock levels, customer orders, deliveries, and inventory transactions. In any business organization, proper stock management is important because it ensures that products are available whenever customers require them. Manual stock management often leads to errors such as incorrect stock counts, delayed deliveries, loss of records, and difficulty in tracking inventory movement. To overcome these issues, a computerized Stock Maintenance System is developed.

The proposed system allows customers to place orders for products through a stock dealer. The stock dealer communicates with the central stock system to verify the availability of products. Once the order is approved, the requested items are delivered to the customer, and the stock database is automatically updated. The system provides better inventory control, reduces human errors, and improves overall operational efficiency.

2. Problem Statement

Many organizations still maintain stock records manually, which can result in inaccurate inventory information, delayed order processing, and poor customer service. When stock details are not updated regularly, organizations may face stock shortages or excess inventory. Therefore, there is a need for an automated Stock Maintenance System that can manage inventory efficiently, process customer orders quickly, and provide real-time stock information.

3. Objectives of the Stock Maintenance System

The major objectives of the Stock Maintenance System are:

3.1 Inventory Management

To maintain accurate records of all products available in the inventory and monitor stock levels continuously.

3.2 Order Processing

To process customer orders efficiently by verifying product availability before confirming orders.

3.3 Stock Verification

To ensure that sufficient stock is available before approving customer requests.

3.4 Automatic Stock Updates

To update inventory records automatically whenever products are sold or delivered.

3.5 Report Generation

To generate reports regarding stock availability, sales transactions, and inventory movement.

3.6 Customer Satisfaction

To improve customer satisfaction by ensuring timely order processing and delivery.

4. Scope of the System

The Stock Maintenance System covers all major inventory-related activities within an organization. The scope includes:

- Product inventory management.
- Customer order management.
- Stock verification and approval.
- Product delivery management.
- Inventory updates.
- Sales and stock reporting.
- Inventory tracking and monitoring.

The system can be used in retail stores, warehouses, manufacturing companies, supermarkets, and distribution centers.

5. Actors in the System

5.1 Customer

The customer is the person who purchases products from the organization. The customer initiates the process by placing an order for required items. Customers expect quick order processing and timely delivery.

Responsibilities:

- Place orders.
- View order status.
- Receive delivered products.

5.2 Stock Dealer

The stock dealer acts as an intermediary between the customer and the inventory system. The dealer receives customer requests, verifies details, and coordinates order processing.

Responsibilities:

- Receive customer orders.
- Verify order information.
- Forward requests to the central stock system.
- Deliver products to customers.

5.3 Central Stock System

The Central Stock System is the core component responsible for inventory management. It stores product information and performs stock verification.

Responsibilities:

- Maintain stock records.
- Verify stock availability.
- Approve or reject orders.
- Update inventory after sales.
- Generate reports.

6. Functional Requirements

The Stock Maintenance System must perform the following functions:

6.1 Customer Management

- Store customer information.
- Track customer purchase history.

6.2 Product Management

- Add new products.
- Modify product details.
- Remove obsolete products.

6.3 Stock Verification

- Check stock availability.
- Verify inventory levels.

6.4 Order Management

- Accept customer orders.
- Process approved orders.
- Track order status.

6.5 Delivery Management

- Arrange product delivery.
- Confirm successful delivery.

6.6 Report Generation

- Generate stock reports.
- Generate sales reports.
- Generate inventory reports.

7. Use Case Diagram Description

The Use Case Diagram illustrates the interaction between actors and the system.

Main Use Cases

1. Place Order
2. Verify Order
3. Check Stock Availability
4. Process Order
5. Deliver Product
6. Update Stock
7. Generate Reports

Relationships

- "Verify Order" includes "Check Stock Availability."
- "Process Order" extends from successful verification.
- "Update Stock" occurs after delivery completion.

8. Use Case Scenario

Use Case Name:

Place Order

Primary Actor:

Customer

Preconditions:

- Customer is registered.
- Products exist in inventory.

Main Flow:

1. Customer selects products.
2. Customer places an order.
3. Stock dealer receives order.
4. Central stock system checks stock availability.
5. Order is approved.
6. Product is delivered.
7. Stock records are updated.

Post Conditions:

- Order is completed.
- Inventory is updated.

9. Class Diagram Explanation

The Class Diagram represents the static structure of the system.

Customer Class

Attributes:

- CustomerID

- CustomerName
- Address
- ContactNumber

Methods:

- placeOrder()
- viewOrderStatus()

Order Class

Attributes:

- Order ID
- OrderDate
- Quantity
- Total Amount

Methods:

- createOrder()
- verifyOrder()

Product Class

Attributes:

- ProductID
- Product Name
- Price
- Stock Quantity

Methods:

- Check Availability()
- Update Quantity()

Stock Dealer Class

Attributes:

- Dealer ID
- Dealer Name

Methods:

- Process Order()
- Deliver Product()

Central Stock System Class

Attributes:

- Inventory Records
- Product Database

Methods:

- Verify Stock()
- Update Stock()
- Generate Report()

10. Sequence Diagram Explanation

The Sequence Diagram shows how objects communicate over time.

Steps:

1. Customer sends order request.
2. Stock Dealer receives request.
3. Dealer forwards request to Central Stock System.
4. System checks inventory database.
5. System verifies stock availability.
6. Approval message is sent.
7. Dealer processes order.
8. Product is delivered.
9. Inventory records are updated.
10. Confirmation is sent to customer.

This sequence ensures proper order processing and inventory control.

Sequence Diagram

Customer -> Stock Dealer : Place Order

Stock Dealer -> Central Stock System : Verify Order

Central Stock System -> Item : Check Availability

Item -> Central Stock System : Stock Status

Central Stock System -> Stock Dealer : Order Approved

Stock Dealer -> Customer : Confirm Order

Stock Dealer -> Customer : Deliver Item

Central Stock System -> Item : Update Stock

Item -> Central Stock System : Stock Updated

Activity Diagram

Start



Place Order



Verify Order



Check Stock Availability



Stock Available?



No Yes



Reject Process Order

Order |



Deliver Item



Update Stock



End

Conclusion

The UML model of the Stock Maintenance System helps manage customer orders, verify stock availability, process deliveries, and update stock records efficiently. The system ensures accurate inventory management and smooth order processing.

UML Design for Online Course Reservation System

1. Introduction

An Online Course Reservation System is a web-based application that allows students to browse available courses and reserve seats in their preferred courses. The entire process is managed by a Central Management System, which maintains course information, seat availability, student records, and enrollment details. The system simplifies the course registration process by allowing students to select courses online instead of using manual registration methods.

In a traditional system, students often face difficulties such as long waiting times, manual paperwork, and lack of information regarding seat availability. The Online Course Reservation System overcomes these issues by automating course selection, seat verification, and enrollment processes. The system ensures transparency, efficiency, and accurate management of course registrations.

2. Problem Statement

Universities offering multiple courses need an efficient mechanism to manage course registrations. Manual registration systems may lead to errors, duplicate entries, and difficulty in tracking available seats. Therefore, an Online Course Reservation System is required to automate the registration process, verify seat availability, and ensure proper enrollment of students.

3. Objectives of the System

3.1 Course Management

To maintain information about all available courses offered by the university.

3.2 Student Registration

To allow students to browse and select desired courses online.

3.3 Seat Verification

To verify the availability of seats before enrollment.

3.4 Enrollment Management

To enroll students automatically when seats are available.

3.5 Database Maintenance

To maintain records of students, courses, and enrollments.

3.6 Report Generation

To generate reports regarding course registrations and seat occupancy.

4. Scope of the System

The Online Course Reservation System includes:

- Student registration and login.
- Course browsing and searching.
- Course selection.
- Seat availability checking.
- Student enrollment.
- Course record maintenance.
- Report generation.
- Centralized course management.

The system can be used by universities, colleges, training institutes, and educational organizations.

5. Actors in the System

5.1 Student

The student is the primary user of the system. Students browse available courses and select courses according to their interests and eligibility.

Responsibilities

- View available courses.
- Select desired course.
- Submit enrollment request.
- View enrollment confirmation.

5.2 University

The university is responsible for maintaining course information and verifying enrollment requests.

Responsibilities

- Manage course offerings.
- Verify seat availability.
- Approve student enrollment.
- Maintain academic records.

5.3 Central Management System

The Central Management System controls all operations related to course reservations.

Responsibilities

- Store course details.
- Check seat availability.
- Process enrollment requests.
- Update seat counts.
- Generate reports.

6. Functional Requirements

The system should perform the following functions:

Student Management

- Register students.
- Maintain student profiles.
- Track course enrollments.

Course Management

- Add new courses.
- Update course information.
- Remove obsolete courses.

Reservation Management

- Accept course reservation requests.
- Verify seat availability.
- Confirm enrollment.

Seat Management

- Monitor available seats.
- Update seat counts automatically.

Report Management

- Generate enrollment reports.
- Generate course occupancy reports.

7. Use Case Diagram Description

Actors

- Student
- University
- Central Management System

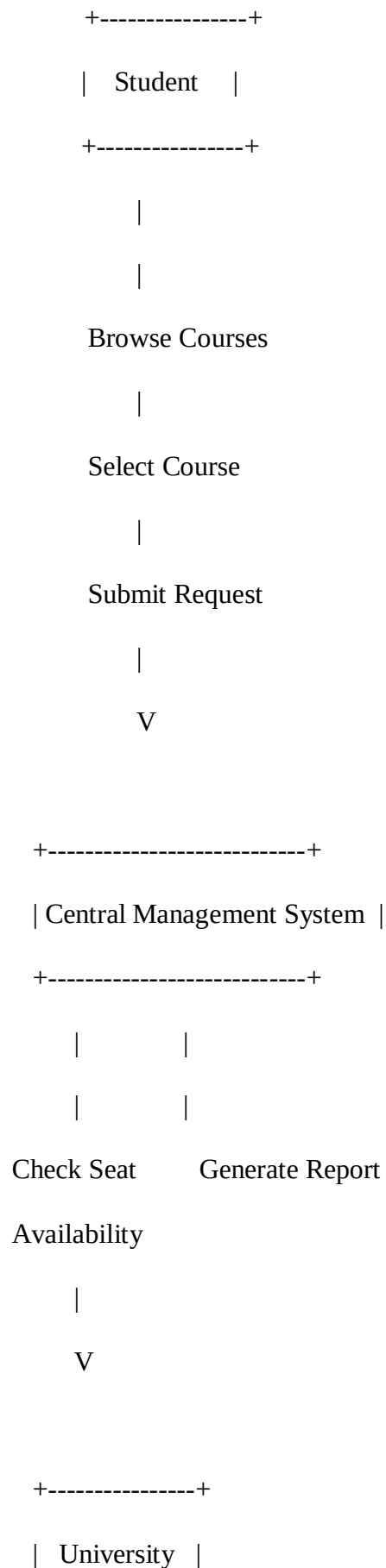
Use Cases

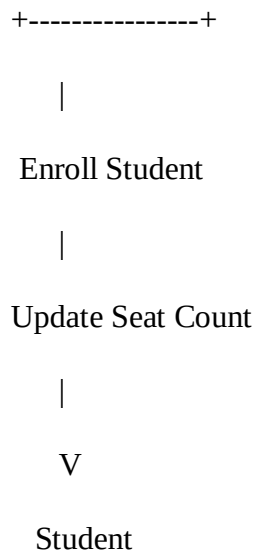
1. Browse Courses
2. Select Course
3. Submit Reservation Request
4. Check Seat Availability
5. Enroll Student
6. Update Seat Count
7. Generate Reports

Relationships

- "Select Course" includes "Browse Courses."
- "Enroll Student" includes "Check Seat Availability."
- "Update Seat Count" occurs after successful enrollment.
- "Generate Reports" is performed by the Central Management System.

8. UML Use Case Diagram (Text Representation)





Applications of Online Course Reservation System

- Universities
- Colleges
- Online Learning Platforms
- Training Institutes
- Professional Certification Programs
- Distance Education Centers

Benefits of the System

Efficient Course Registration

Students can register for courses quickly without visiting administrative offices.

Real-Time Seat Information

Students can view current seat availability before selecting a course.

Reduced Administrative Work

Automates enrollment procedures and reduces paperwork.

Accurate Data Management

Maintains secure and accurate records of students and courses.

Conclusion

The Online Course Reservation System is an effective solution for managing course registrations in educational institutions. It allows students to browse and select courses, verifies seat availability, and performs automatic enrollment when seats are available. UML diagrams such as Use Case Diagram, Class Diagram, Sequence Diagram, and Activity Diagram help in understanding and designing the system efficiently. The system improves registration accuracy, reduces administrative workload, and enhances the overall learning experience for student

Code of Conduct:

I K Madan Mohan Reddy (192411206) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Kyreddy

RUBRICS FOR CALCULATION:**Experiments**

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation